

清华大学基础模型2025学术年会

分论坛六 – 具身智能

具身智能的自身模型：运动-神经系统的动力学建模与控制
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Self Model for Embodied Intelligence

Modeling and Control of Human Neuro-Musculo-Skeletal System

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YananSui.com

Neuro-Muscle-Skeletal Multiphysics

Modeling and Control



Nervous System



Neural Stimulation



Muscle-Skeletal Motion

Self Model for Embodied Intelligence

model

control

emotion

dynamics



Embodied Intelligence: Learning to Move

neuro-musculo-skeletal modeling
and reinforcement learning
for human motion control and interactive robotics

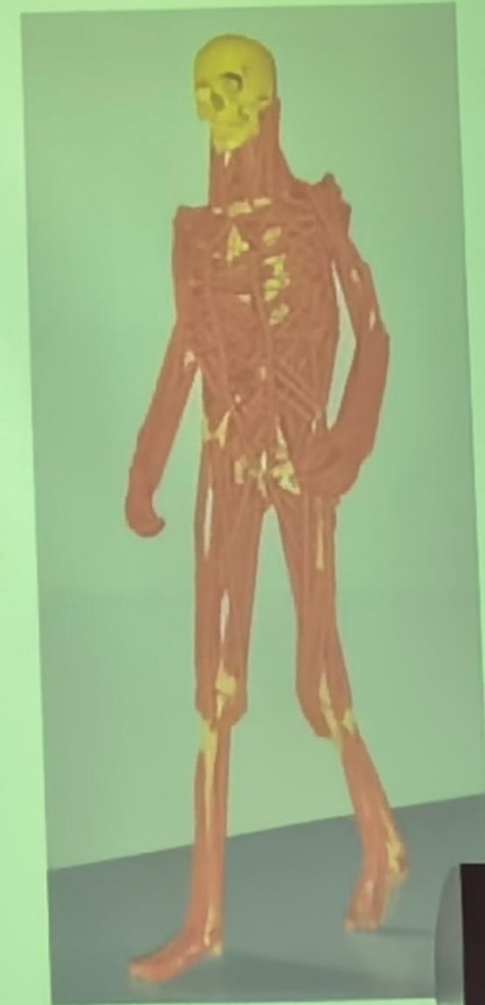
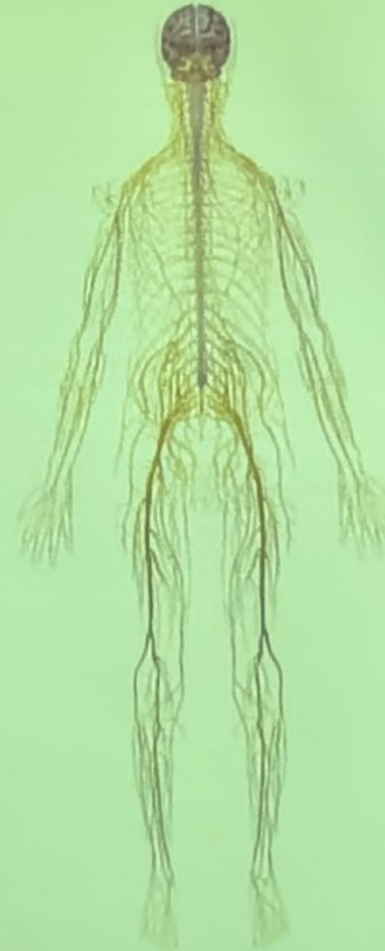
optimization with

- 1. safety constraints**
- 2. preference feedback**
- 3. structured decision spaces (models)**

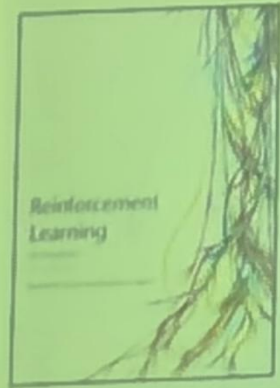


Embodied Intelligence: Learning to Move

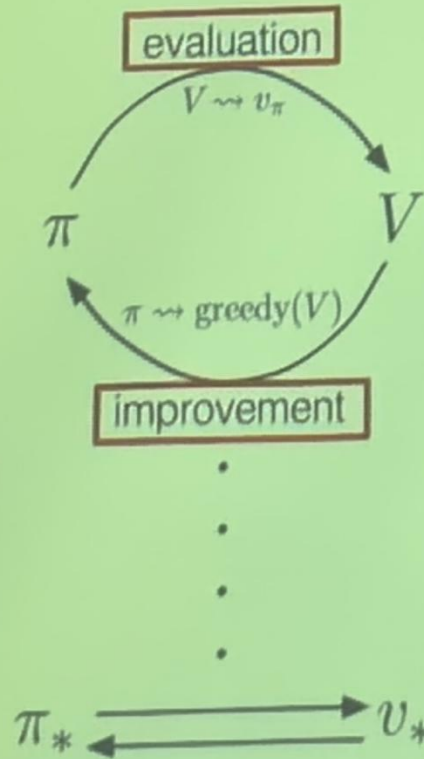
- Human motor functions
 - › ~100 billion neurons in the brain
 - › ~150,000 motor neurons control the contraction of over 600 muscles
- How to control?
 - › A biological problem
 - › Also a computational problem
- Reinforcement learning
 - › Online <> Offline
 - › Model-free <> Model-based



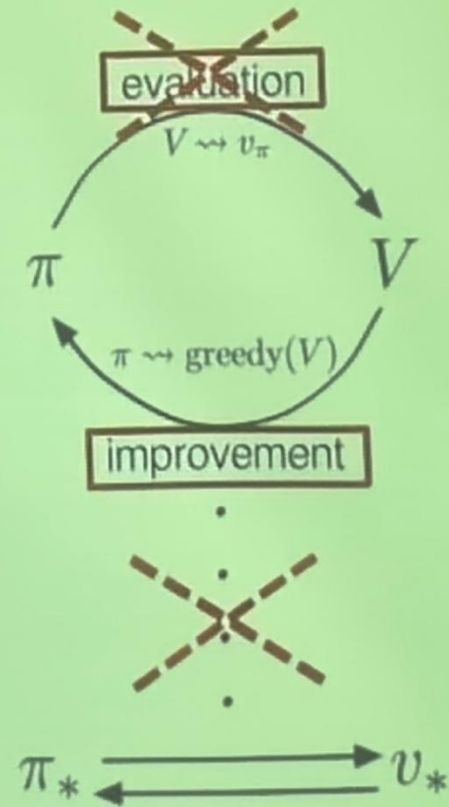
Safe Reinforcement Learning



Sutton & Barto



**Unknown
Safe
Constraints**



**Unknown Safety Constraints
Break the Reinforcement Learning Loop**

Safe Optimization & Reinforcement Learning

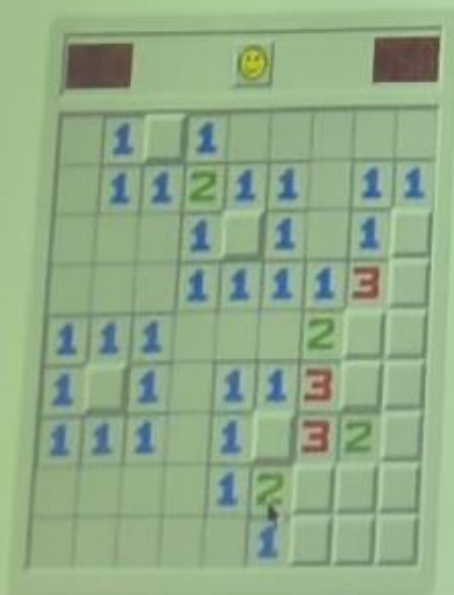
- Safe Optimization Algorithms

- › Use posterior to determine safe region
- › Safety guarantee with high probability

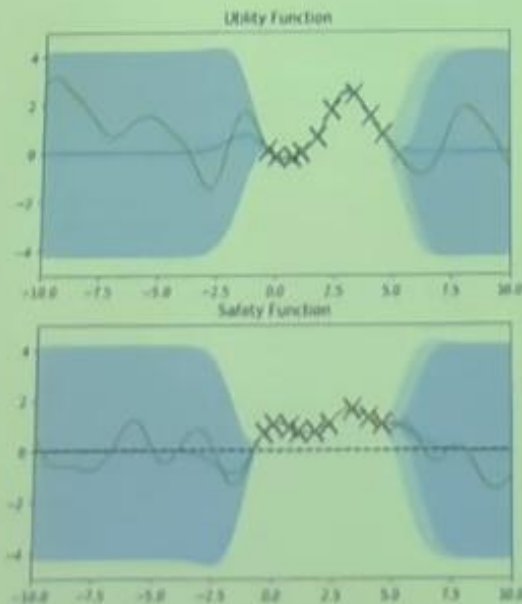
Safe Expansion



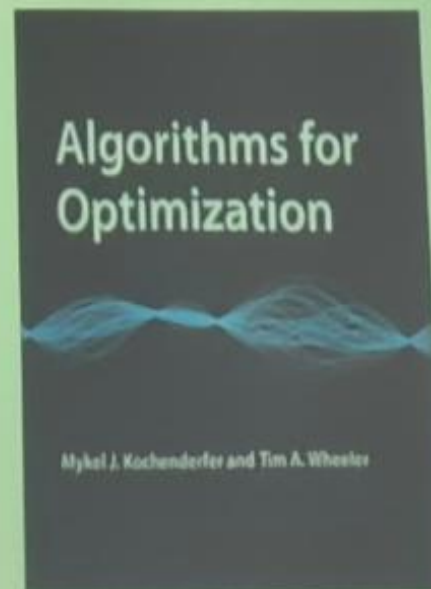
Exploration ↔ Exploitation



Minesweeper



SafeOpt Procedure



Chapter 16.6

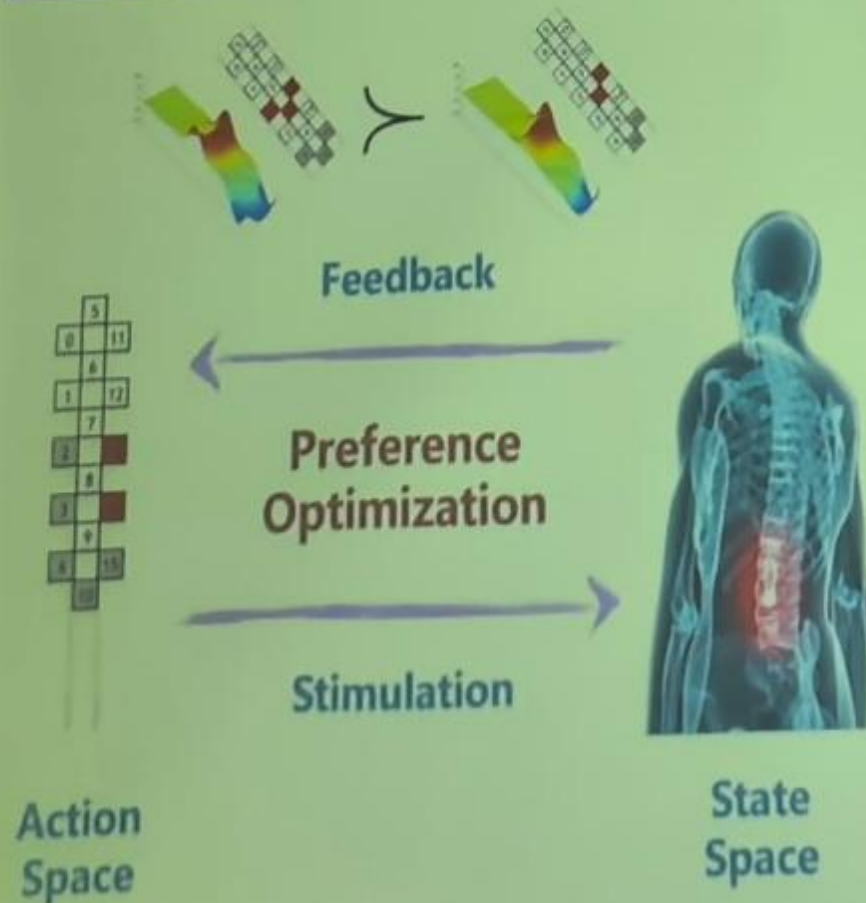
Sui et al., ICML'15, ICML'18

Wachi & Sui, ICML'20

Wachi et al., NeurIPS'21

Wei et al., CoRL'24

Preference-based Optimization: from the inside out



User Preference Feedback



Sui & Burdick. Clinical online recommendation with subgroup rank feedback. RecSys(2014)
Sui, Yue, Burdick. Correlational dueling bandits with application to clinical treatment in large decision spaces. IJCAI(2014)
Sui et al., Stagewise Safe Bayesian Optimization with Gaussian Process. NIPS(2015)

Preference-based Optimization: from the outside in



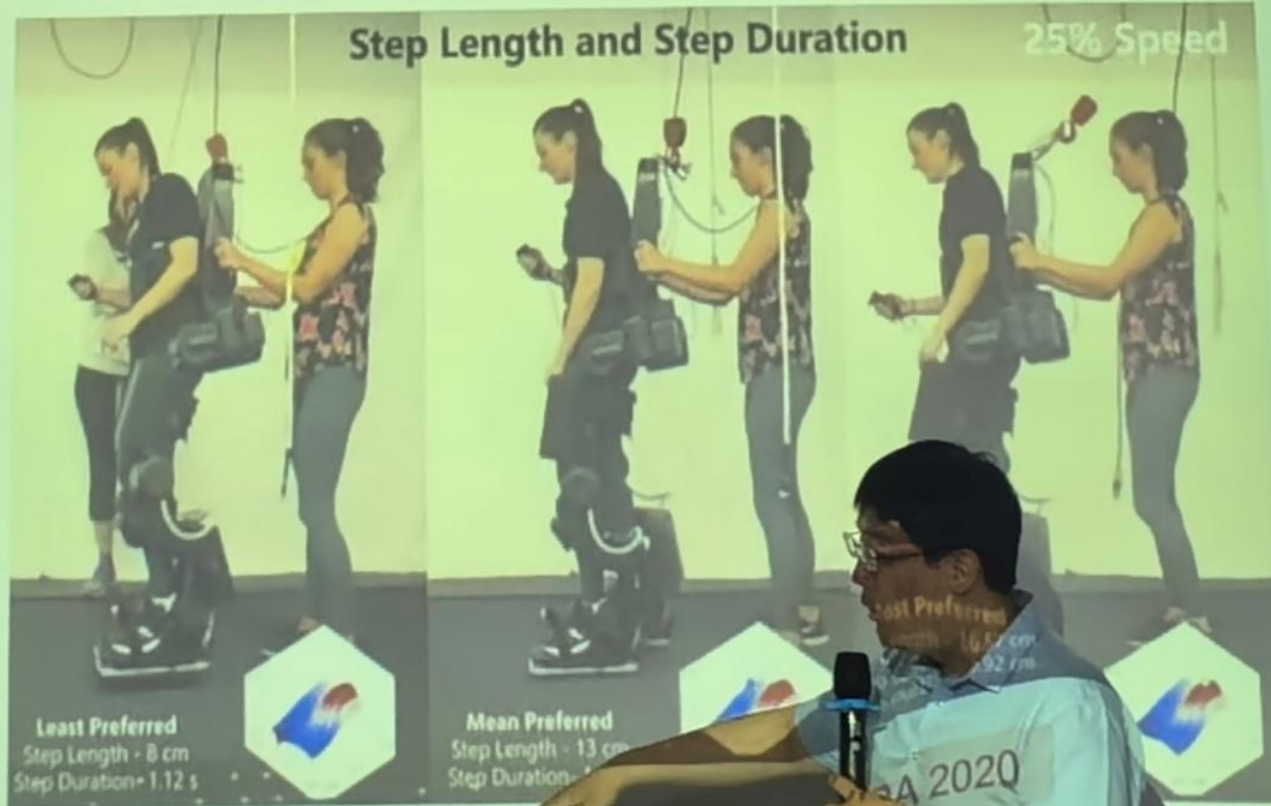
Algorithm 4 SelfSparring

```

1:  $TS \leftarrow$  new Thompson Sampling SBM
2:  $TS.init(), t \leftarrow 0$ 
3: while  $t \geq 0$  do
4:    $i_1 = TS.queryAction()$ 
5:    $i_2 = TS.queryAction()$ 
6:   play  $i_1, i_2$ , observe rewards  $r_1, r_2$ 
7:    $TS.feedback(i_1, r_1)$ 
8:    $TS.feedback(i_2, r_2)$ 
9:    $t \leftarrow t + 1$ 
10: end while
    
```

Theorem 1. Under Approximate Linearity, INDSELF-SPARRING converges to the optimal arm b_1 as running time $t \rightarrow \infty$: $\lim_{t \rightarrow \infty} \mathbb{P}(b_t = b_1) = 1$.

Convergence with
Self-Spar (Self-Play)



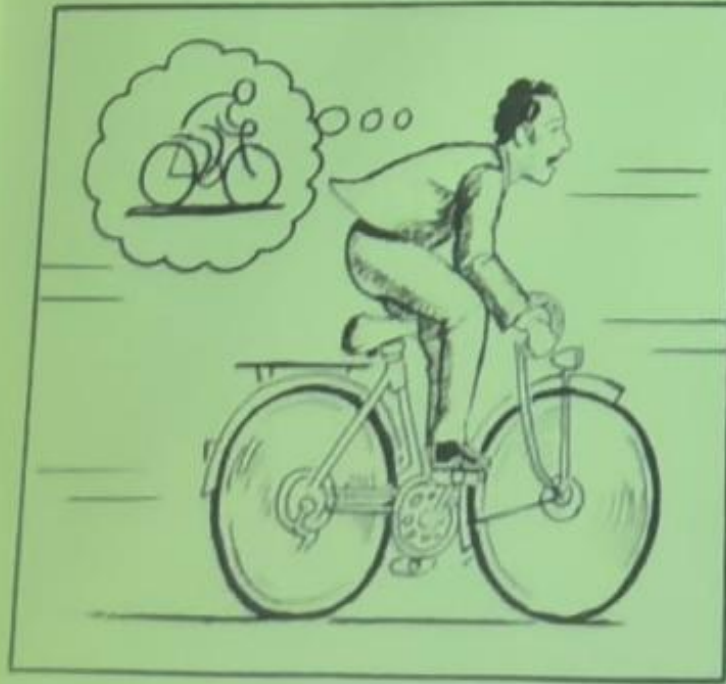
Best Conference Paper Award, ICRA 2020

Best Paper Award on Human-Robot Interaction, ICRA 2020

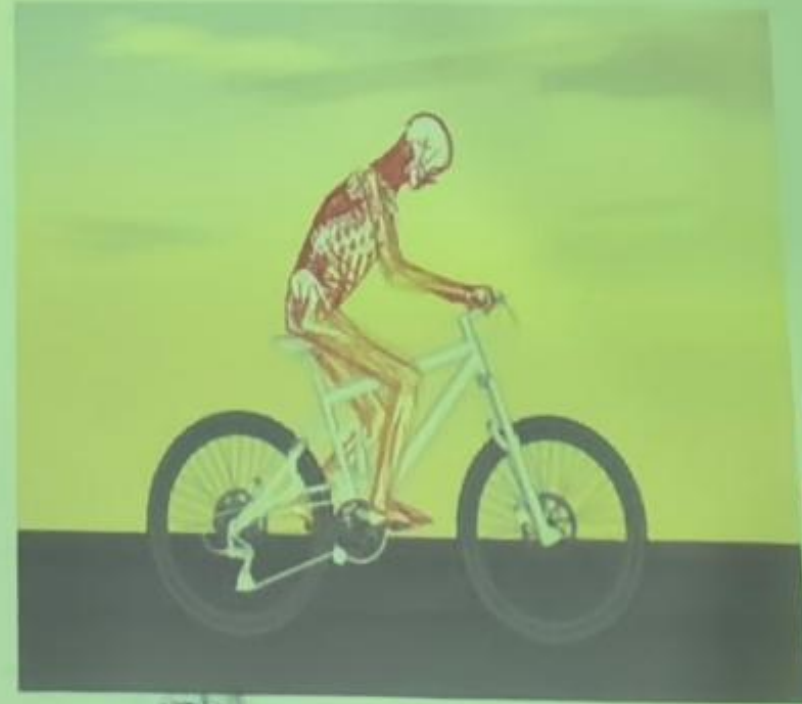
World Model

The image of the world around us, which we carry in our head, is just a model.

(Forrester, 1971)



Abstraction of the World



Abstraction of the World

Self Model

Know Thyself

(Delphic maxim)



NVIDIA Project GROOT



SPML Graphics Model



CGI



MS-Human 700



Whole-body Natural Walking

Ours

E2E

Sar-E2E

Sar-Ours

Training Steps: 15M+15M



Walking
Forward



Human-Exo
Interaction

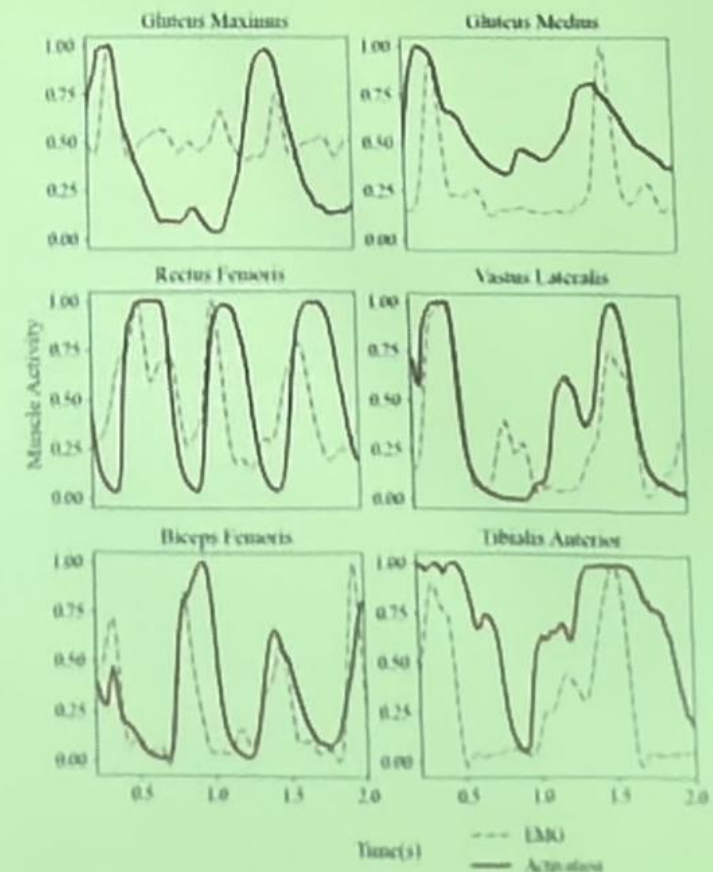
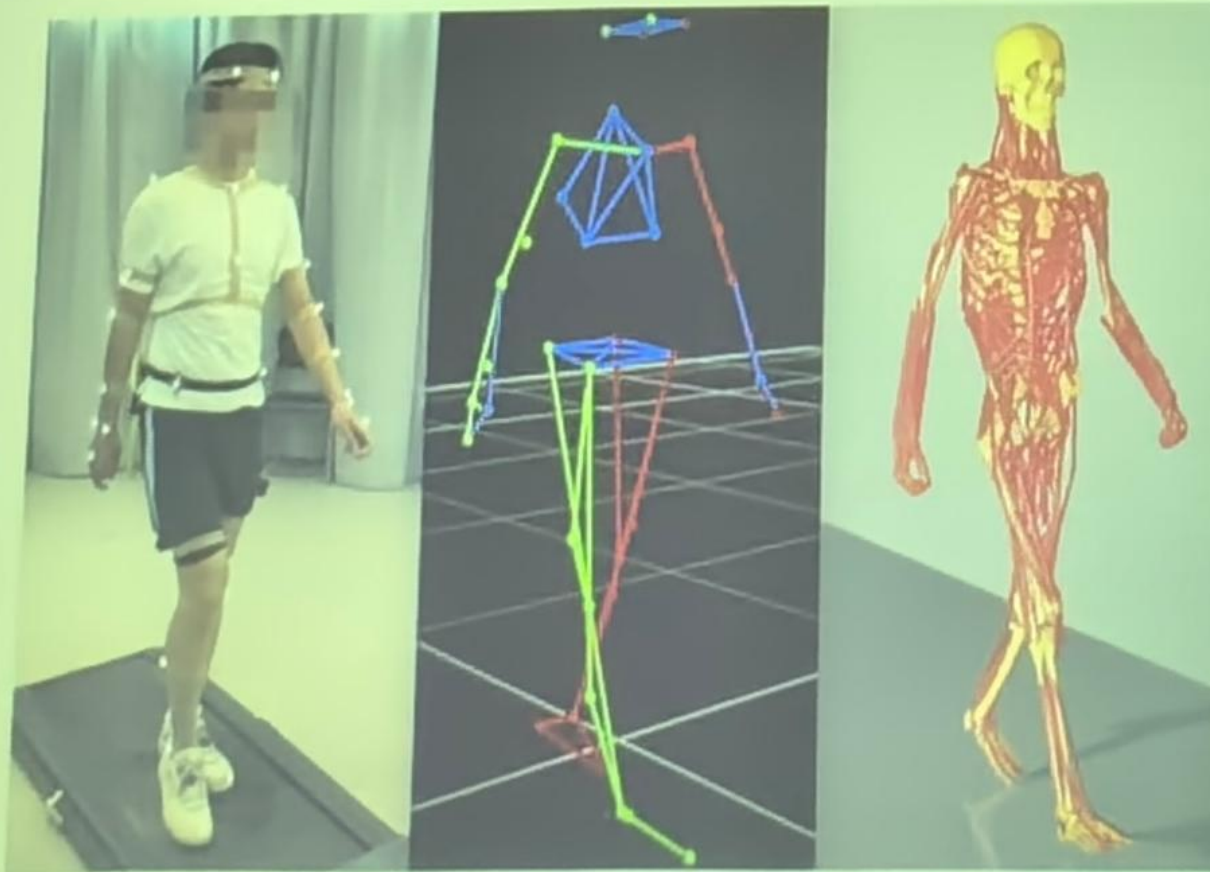


Prosthetic
Walking



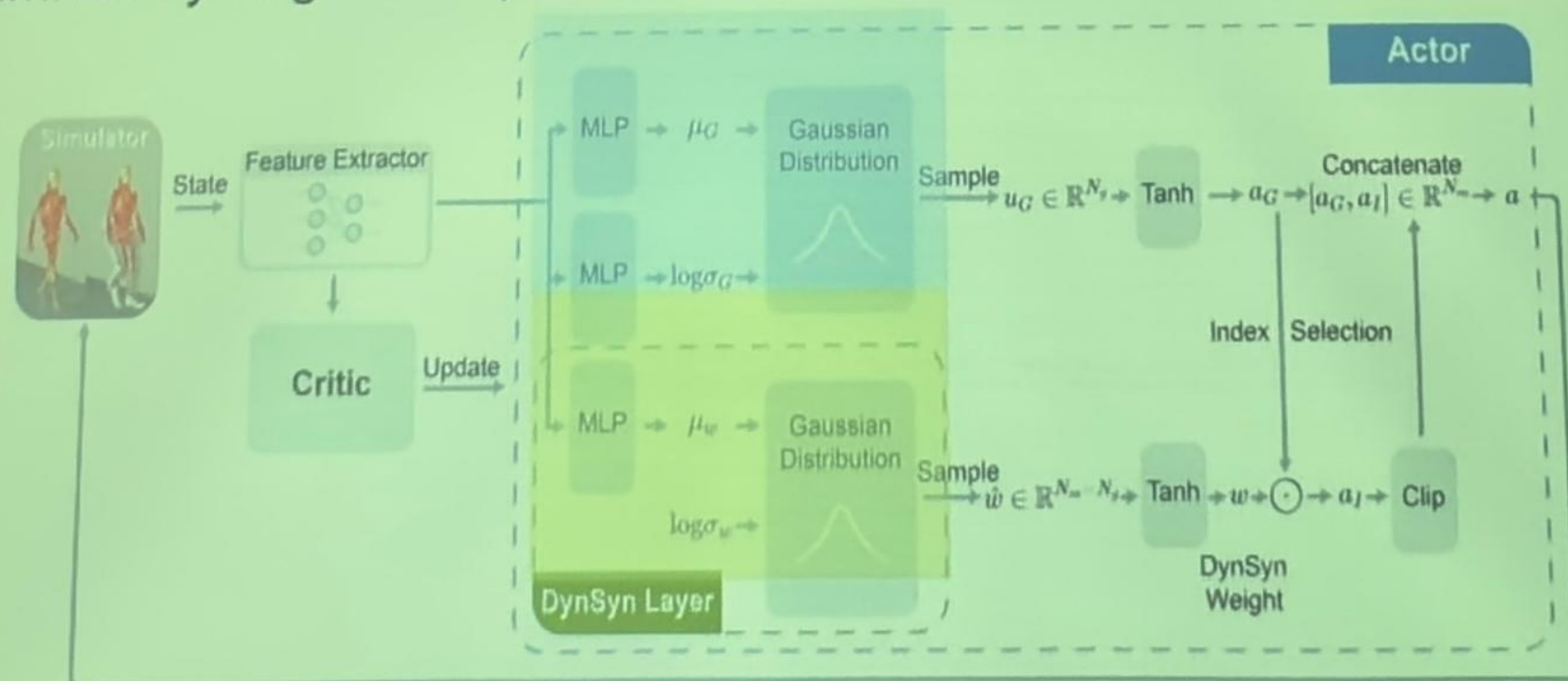
Full-body Human Motion Control

- Model Authenticity Verification



Efficient Control in Overactuated Embodied Systems

- Dynamical Synergistic Representation for Efficient Learning



$$w = \tanh(\hat{w}), \hat{w} \sim \mathcal{N}(\mu_w, \sigma_w)$$

$$c = \min(\max(k_D t + a_D, 0), \kappa)$$

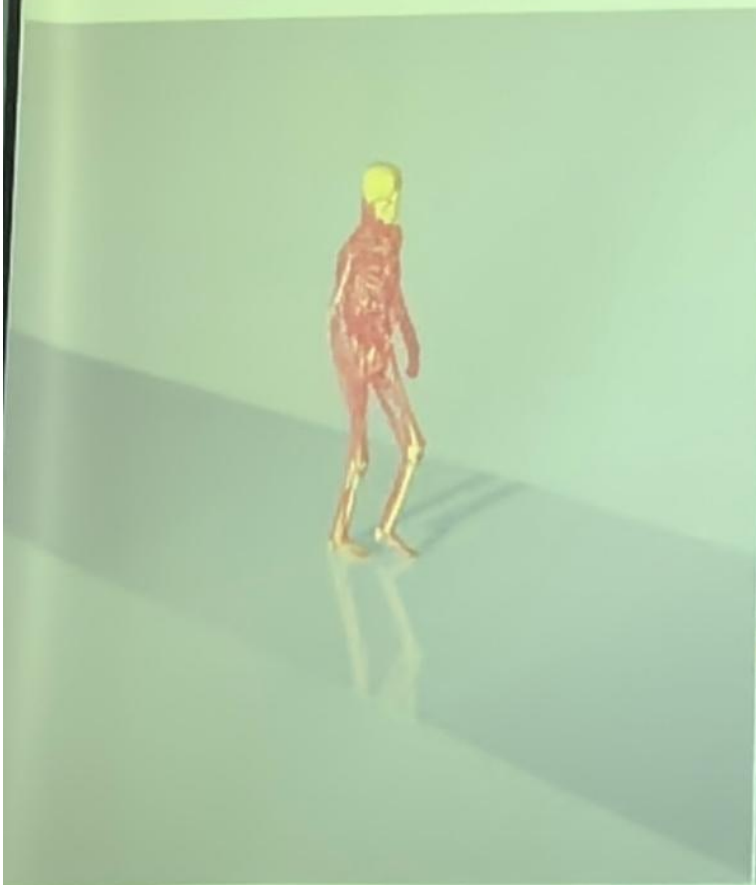
$$a_I = \text{IS}(a_G) \odot \text{clip}(\kappa w, -c, c)$$

$$a_G = \tanh(u_G), u_G \sim \mathcal{N}(\mu_G, \sigma_G)$$

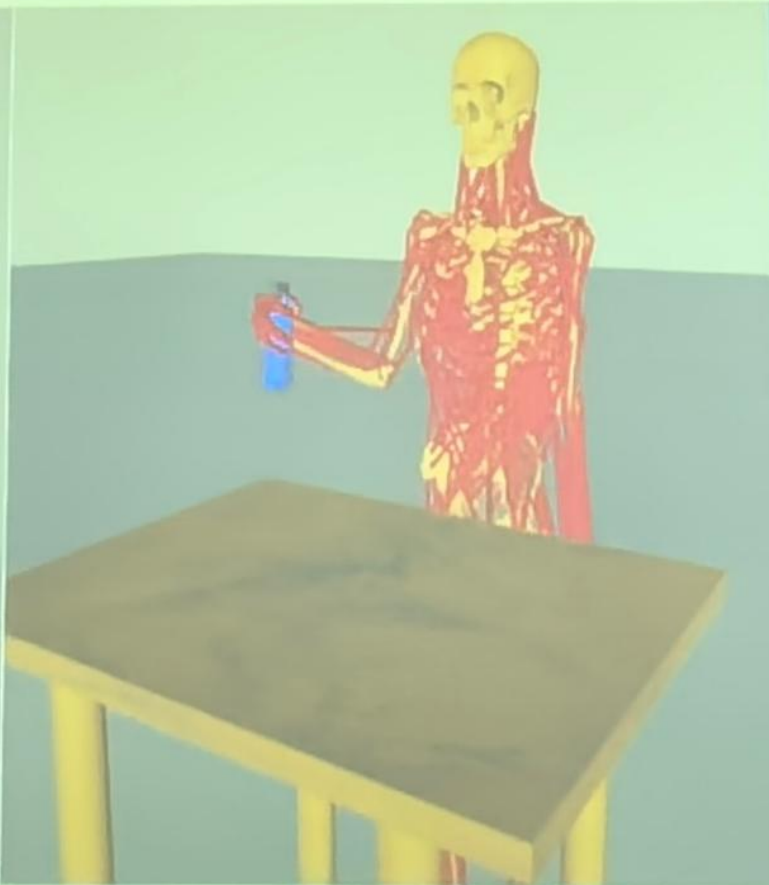
$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t) + \alpha H(\pi(a_G | s_t))) \right]$$

Efficient Control in Overactuated Embodied Systems

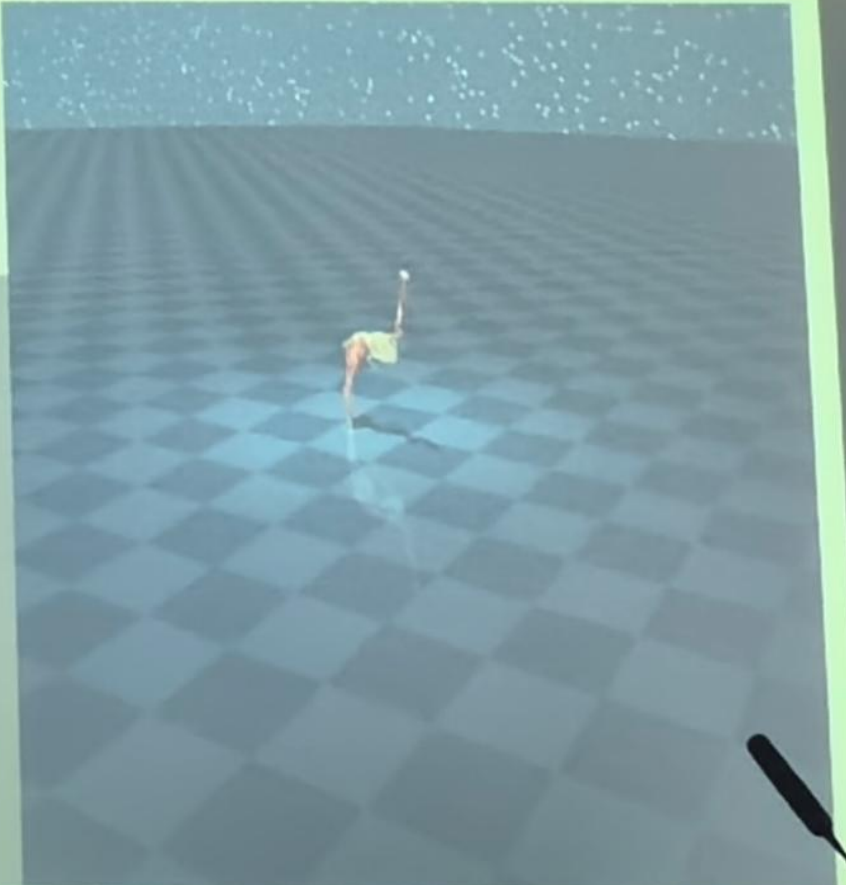
- Dynamical Synergistic Representation for Efficient Learning



MS-Human-700-Gait



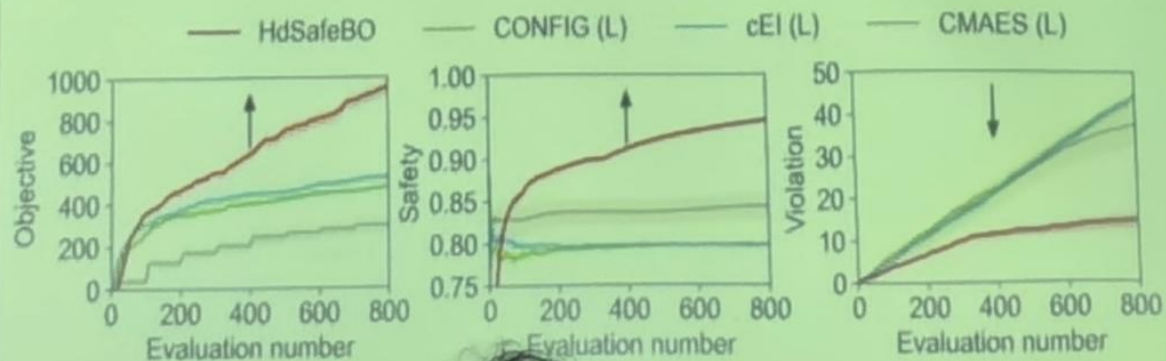
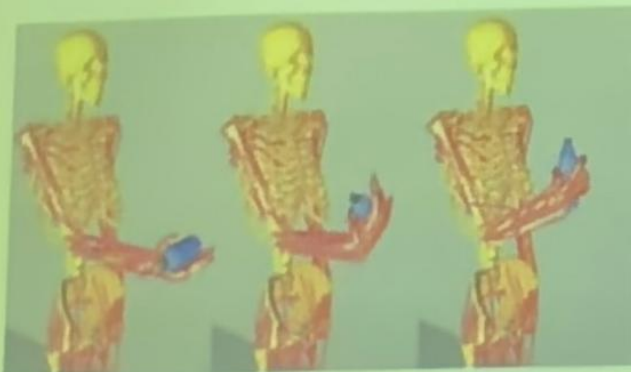
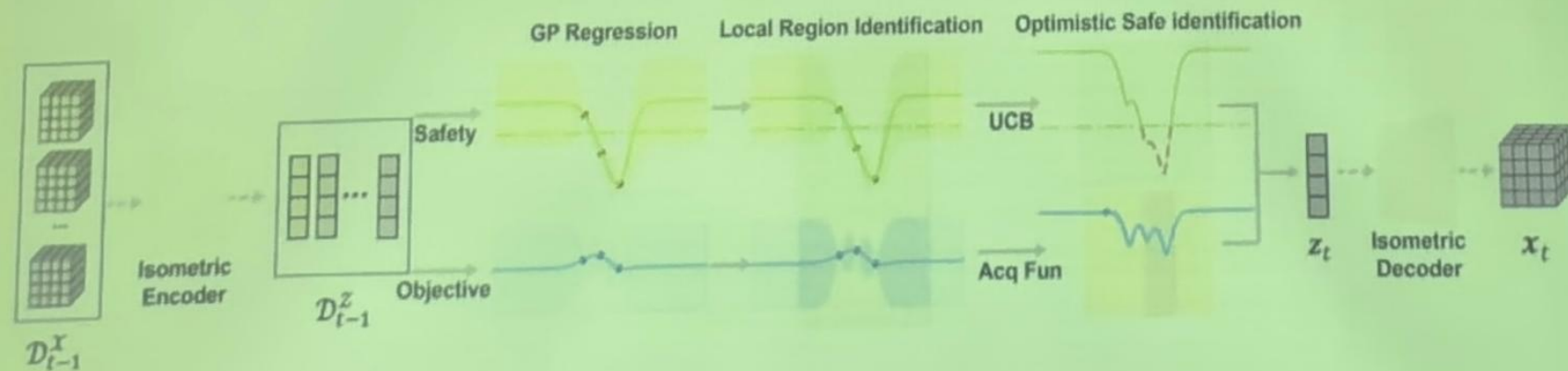
MS-Human-700-Locate



Ostrich-Run

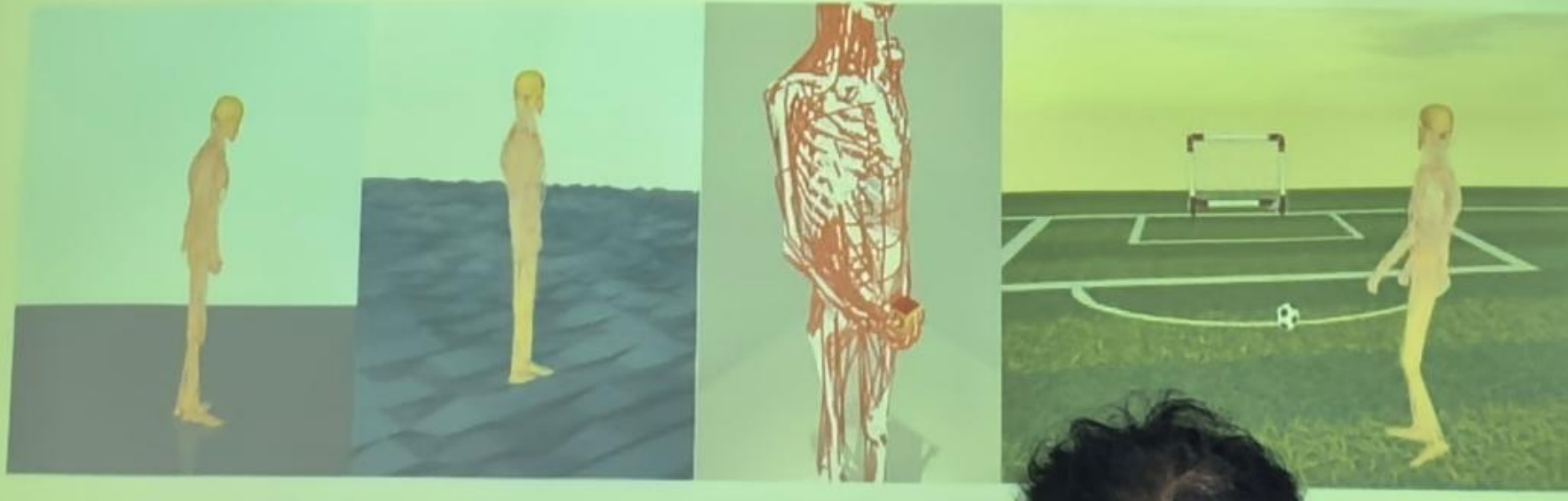
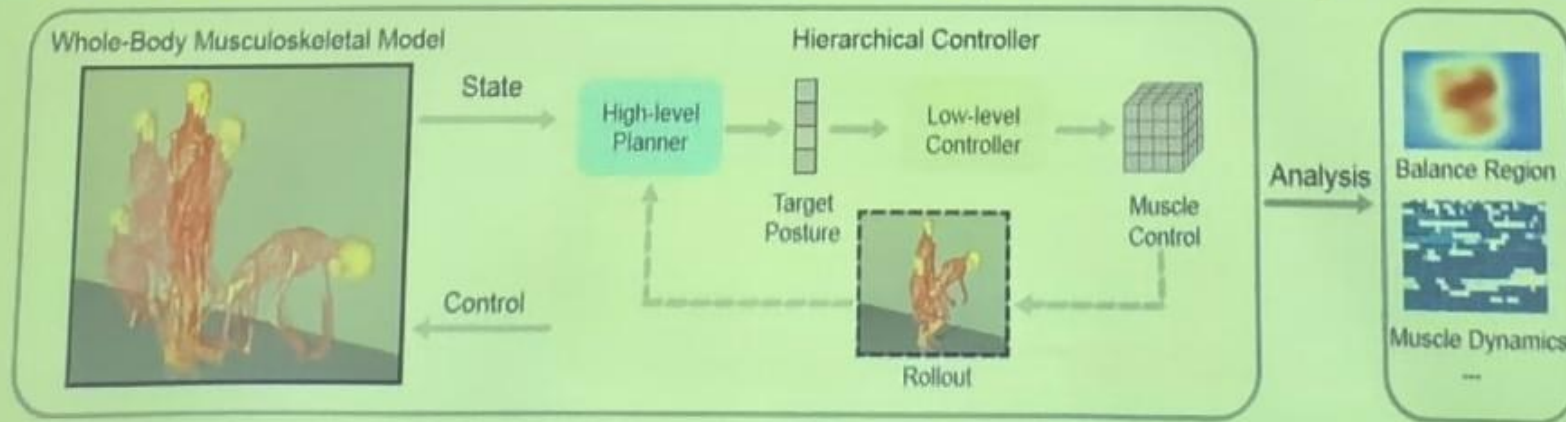
Safe Control in Overactuated Embodied Systems

- Safe Bayesian Optimization for High-Dimensional Control

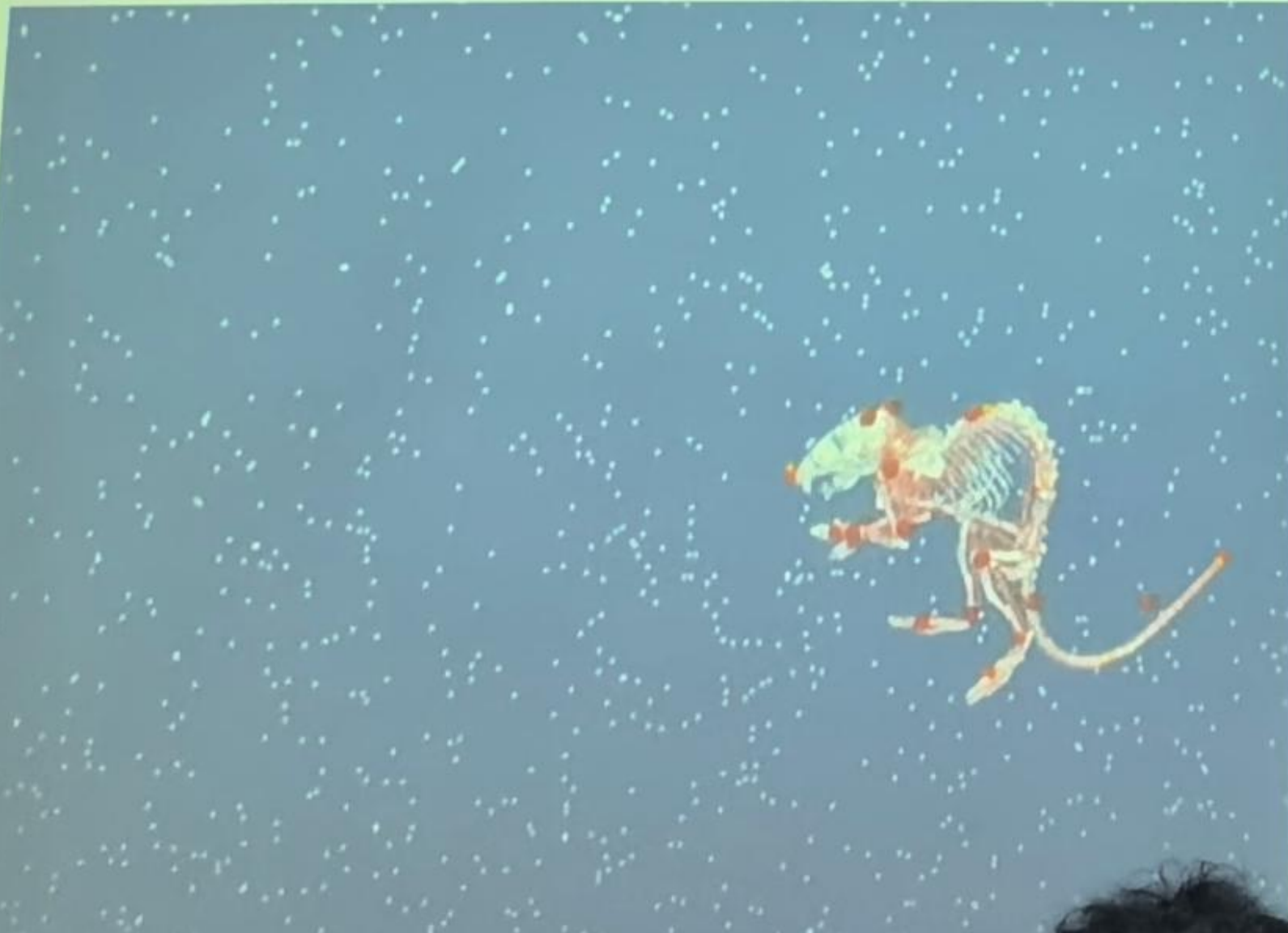


Hierarchical Model-Based Planning

- Model Predictive control with Morphology-aware Proportional Control



Neuro-Muscle-Skeletal Modeling and Control



Summary: Learning to Move

neuro-musculo-skeletal modeling and reinforcement learning
for human motion control and interactive robotics

World Model



Self Model



Self Control



谢谢！